

Gaze-Aware Dialog Systems

Integrating Visual Attention for Turn-Taking and Reference Resolution

Dr. Fares Abawi

W3C Workshop on Voice Agents – February 2026

Gaze-Aware Dialog Systems

The Problem: Underutilized Modality

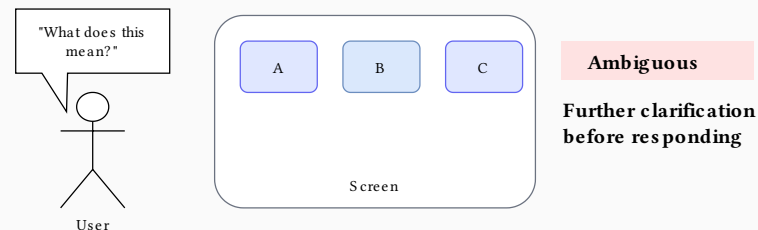
Current Voice Agents:

- Rely solely on acoustic signals
- Miss early conversational cues
- Struggle with ambiguous references
- Reactive turn-taking models

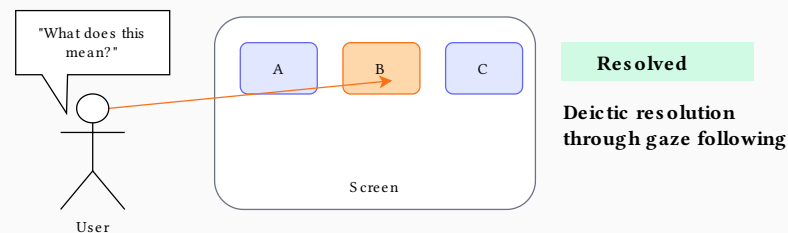
Gaze provides:

- Early intent signals
- Reference grounding (deictic)
- Turn coordination cues

Traditional: Audio-only



Gaze-aware: Multimodal



Use Cases: Where Gaze Matters

Turn-Taking Coordination

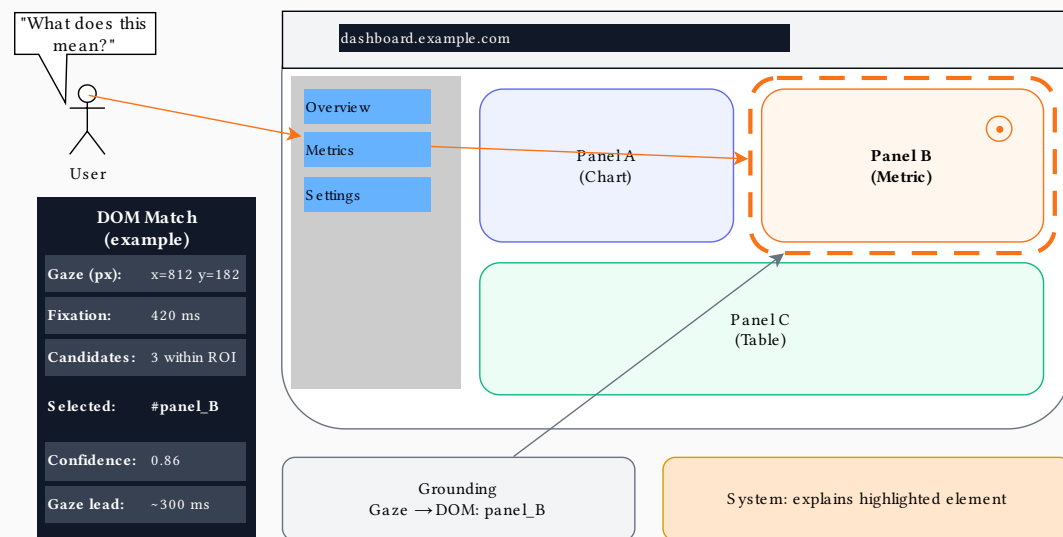
Gaze shifts precede speech onset, enabling anticipatory system states

Barge-In Detection

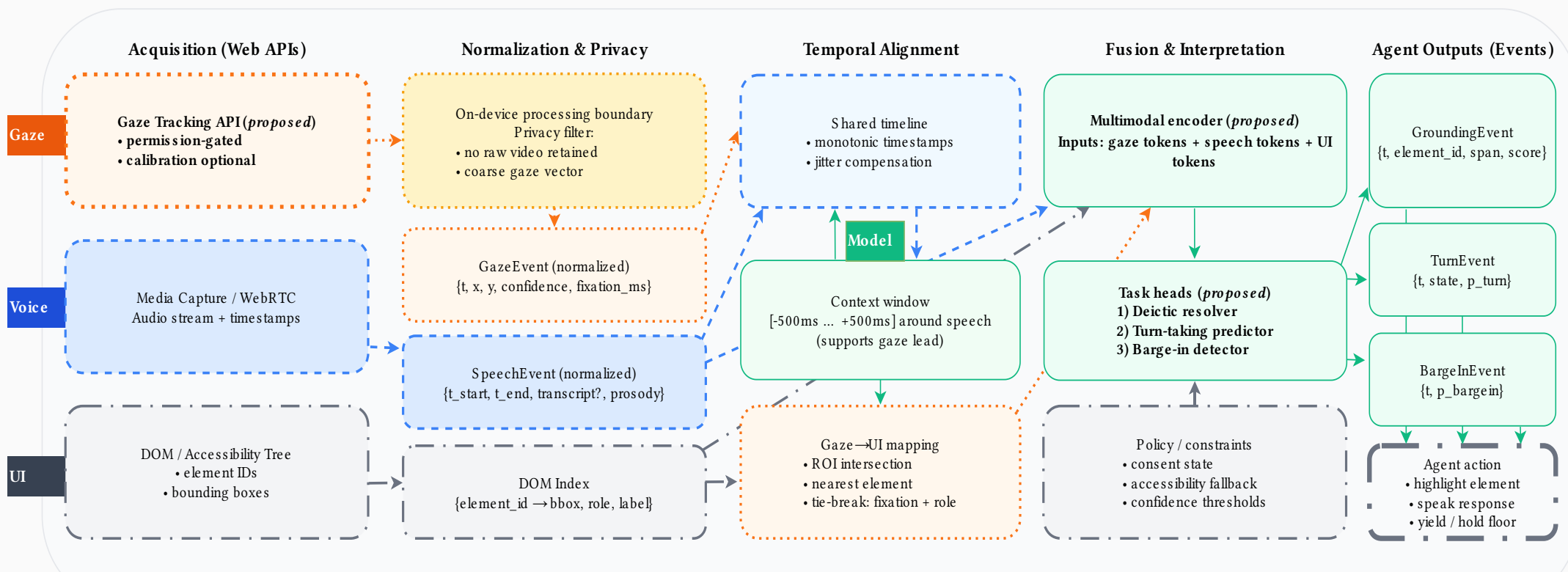
Sustained fixations on interface elements signal interruption intent

Reference Resolution

Direct grounding of deictic expressions via gaze-to-DOM mapping



Multimodal Pipeline Overview



Note: event schemas shown are illustrative; standardization targets include permission model, timestamping, and privacy-preserving gaze representations.

Technical Approach: Multimodal Fusion

Pipeline Architecture — Three parallel input streams unified through temporal alignment

1. Acquisition (Web APIs)

Gaze Tracking API (permission-gated); Media Capture for timestamped audio; DOM/Accessibility Tree for element bounding boxes and roles

2. Normalization & Privacy

On-device processing with no raw video retention; coarsened gaze vectors; unified event schemas for speech and gaze

3. Temporal Alignment

Shared monotonic timeline with jitter compensation; $\pm 500\text{ms}$ context window captures anticipatory gaze preceding speech

4. Fusion & Interpretation

Cross-attention encoder over gaze, speech, and UI tokens; policy layer enforces consent and confidence thresholds

5. Task Heads

Deictic resolver maps gaze-speech to DOM; turn predictor estimates floor state; barge-in detector identifies interruption intent

6. Agent Outputs

Structured events (GroundingEvent, TurnEvent, BargeInEvent) drive element highlighting, responses, and floor management

Neural Integration Mechanisms

Early Fusion

Concatenate gaze vectors with speech and UI embeddings at input level

$$\mathbf{h} = f([\mathbf{g}; \mathbf{s}; \mathbf{u}])$$

Simple; joint learning from start; may lose modality-specific patterns

Late Integration

Separate encoders per modality; combine at decision level

$$\mathbf{y} = \sigma(\mathbf{W} [f_g(\mathbf{g}); f_s(\mathbf{s}); f_u(\mathbf{u})])$$

Preserves modality-specific learning; flexible architecture

Cross-Modal Attention

Gaze queries attend to speech and UI keys; learns temporal alignment

$$\mathbf{h}_g = \text{Attn}(\mathbf{Q}_g, \mathbf{K}_{s,u}, \mathbf{V}_{s,u})$$

Dynamic cross-modal relationships; effective for deictic grounding

Gated Multimodal Units

Learnable gates control modality contribution to the final output

$$\mathbf{h} = \mathbf{z} \odot \tanh(\mathbf{W}_g \mathbf{g}) + (1 - \mathbf{z}) \odot \mathbf{h}_{s,u}$$

Handles missing gaze gracefully; potential for post-training modality replacement; adaptive fusion

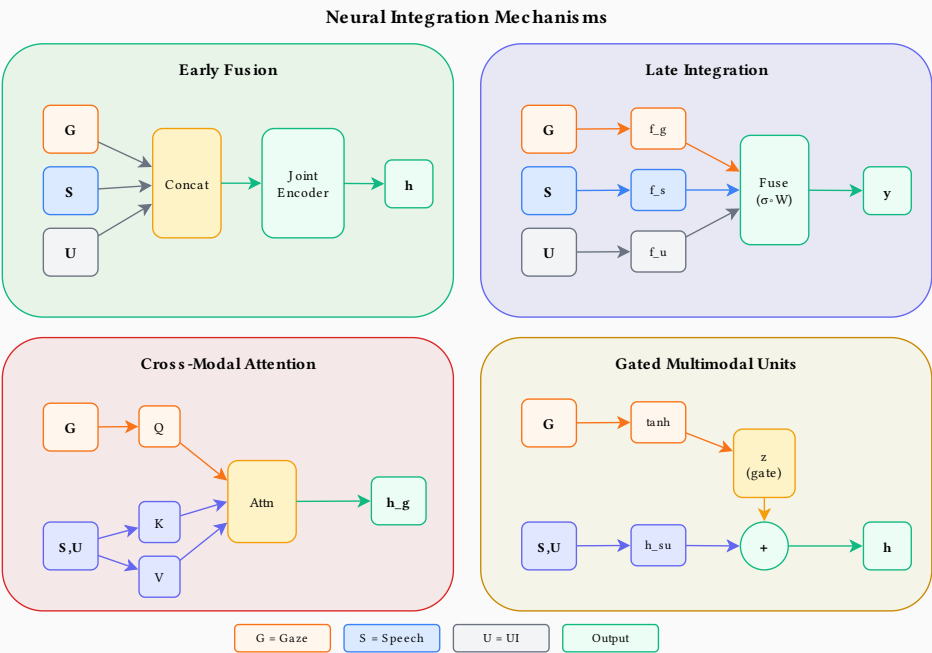
Architecture Comparison

Method	Latency	Robustness	Accuracy
Early	Low	Low	Medium
Late	Medium	High	Medium
Attention	High	Medium	High
Gated	Medium	High	High

Table 1: Comparison of neural fusion methods across latency, robustness to missing modalities, and accuracy metrics

Selection Criteria:

- Real-time constraints → Early/Gated
- Missing modality handling → Gated/Late
- Reference resolution → Attention
- Resource-constrained → Early fusion



Hybrid approaches combining gated units with cross-modal attention show promise for production systems

Implementation Challenges

Technical Constraints:

- **Accuracy:** Webcam gaze 4° error
 - Sufficient for coarse element selection
 - Insufficient for precise text tracking
- **Latency:** Real-time processing requirements
 - <100ms for natural turn-taking
 - Edge vs. cloud computation tradeoffs
- **Robustness:** Variable lighting conditions
 - Head pose variations
 - Multiple monitor setups
 - Off-screen referents

Privacy & Deployment:

- **User consent:** Webcam permissions
 - Transparent data usage policies
 - On-device processing preferred
- **Standardization needs:**
 - Gaze data formats for Web APIs
 - Browser support consistency
 - Cross-platform compatibility
- **Accessibility:** Alternative modalities
 - Graceful degradation without gaze
 - Assistive technology integration

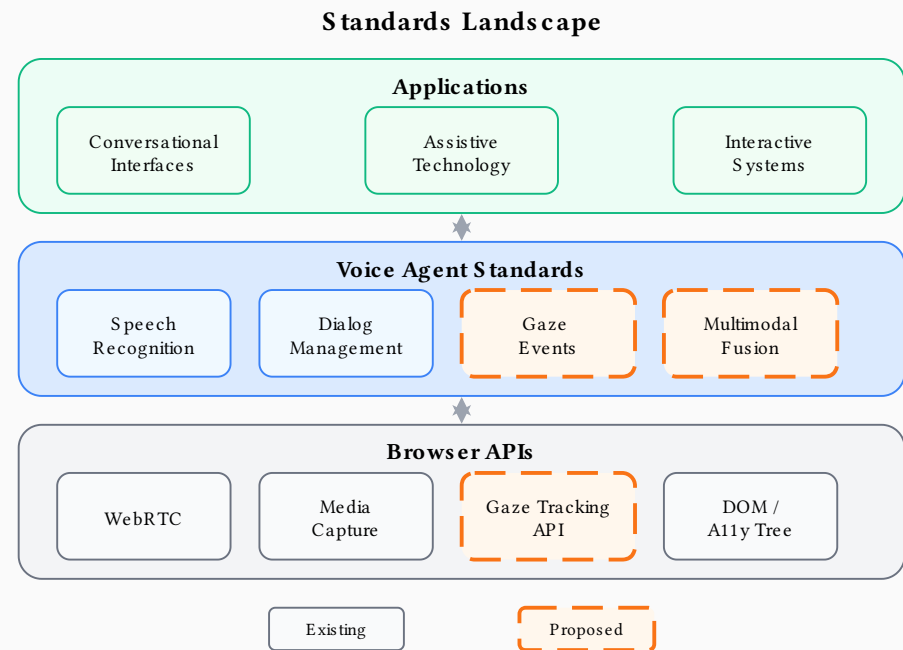
Balancing innovation with practical constraints requires community consensus on standards

Path Forward: Standardization Needs

W3C Standards Integration

Proposed Specifications:

- Gaze Tracking API for web browsers
 - Integration with existing tools (e.g., WebGazer.js)
 - Calibration protocols
 - Privacy-preserving data formats
- Voice Agent Multimodal Extension
 - Multimodal fusion
 - Gaze event integration
 - Reference resolution standards
- Accessibility Guidelines
 - Fallback mechanisms
 - Alternative input methods
 - Gaze-based input for users with motor impairments



Research Questions for Community

- What accuracy thresholds are acceptable?
- How to handle calibration in web contexts?
- Privacy-utility tradeoffs in gaze data?

Dr. Fares Abawi

fares@abawi.me

Let's discuss standardization pathways for gaze-aware voice agents

Slides available at: <https://fares.abawi.me>